

Vulhub AI Tutor

Quick Start & Usage Guide

Before You Start

The full Vulhub lab library (every README + docker-compose file) ships built into this app — nothing to clone, configure, or connect before you can start browsing, reading, and learning. The app never runs Docker or connects to anything on your behalf: each lab page shows you the exact commands to run it yourself, on your own machine or VM. Nothing survives a reboot — the two dev servers need to be started fresh every session (and your VMs too, if you use them). Everything you've done in the app itself (progress, quizzes, notes, VM configuration) is saved in a local SQLite database and will still be there.

- **Required:** nothing extra — just double-click start.bat and go.
- **Optional:** assign a Kali + Vulhub VM in Settings → Virtual Machines for VM power on/off buttons on the dashboard — see the README's "Optional: VM power control" section.

Starting Everything Up

1

Double-click start.bat

In the project root. First run, it also sets up the backend virtual environment and runs npm install for you — after that it just opens two windows and your browser at <http://localhost:5173> in one click. The lab library is already there — nothing else to do.

Running it manually instead? Don't add --reload

start.bat already leaves it off. On this setup --reload breaks the OpenAI SDK's client and every AI call fails with a 500 — restart the backend manually after code changes instead.

2

(Optional) Power on your VMs

Only if you've set up the optional VM integration: start your Kali VM and your Vulhub VM in VMware, either directly or from the app's Settings page.

Using the App

Dashboard

Backend API and AI Tutor status at a glance, plus Kali/Vulhub VM status if you've configured the optional VM integration.

Labs

Browse or search the built-in library — already populated, nothing to scan or configure. **Rescan** just re-reads the bundled library (only useful if you edit it yourself).

Inside a Lab

- **How to Run This Lab** — the lab's path within vulhub, the ports it exposes, and a template docker compose command, for running it yourself on your own Ubuntu VM
- **Explain With AI** — a structured breakdown of the vulnerability
- **Ask the AI Tutor** — contextual Q&A about that specific lab
- **Quiz Me** — a short generated quiz, graded on submit
- **Generate Study Notes** — a saved summary for later review
- Mark your **progress** (not started / in progress / completed) and confidence at the bottom

Progress

Everything you've started or completed, across every session, with quiz scores and categories studied.

Settings

Switch the active AI provider (OpenAI / Anthropic). Optionally assign your Kali + Vulhub VMs and power them on/off.

Shutting Down

- Press Ctrl+C in both terminal windows that start.bat opened (or just close them)
- If you were running a lab on your own VM, stop it there yourself — the app never started it, so it won't stop it either
- Shut down your Kali/Vulhub VMs as you normally would, if you used them

Nothing here is lost between sessions — your progress, quizzes, notes, and VM configuration all live in the backend's local database and pick up right where you left off. The lab library itself ships with the app, so it's always there too.